

# Certified Branch Reductions for the Binary Covering Number $K_2(11, 3)$

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## Abstract

The binary covering number  $K_2(11, 3)$  is recorded in the interval  $15 \leq K_2(11, 3) \leq 16$ . We give a reproducible, certificate-oriented reduction of the unresolved size-15 search. Exact shell and parity-refined Delsarte certificates show that every hypothetical 15-word cover has minimum distance at most 5, total pair-ball overlap at least 1712, and total triple-ball overlap at least 280. Translation and coordinate symmetry then yield a complete 150-case normalized partition. Exact orbit-profile certificates, integer-profile enumeration, checked DRAT traces, and maximum-degree normalization close 112 normalized branches and leave 38 explicit residual branches. Those residual branches contain 2548 third-word orbit children before static exclusions and 2163 live children afterward. In four selected unresolved children, an exhaustive 350-branch fourth-word split has 184 reverse-unit-propagation closures and 140 additional solver-generated DRAT closures. Thus 324 selected branches are certified closed and 26 remain. No complete selected child or normalized parent is closed by this final layer. The exact value and the known interval are unchanged.

## 1 Introduction

A binary code  $C \subseteq \mathbb{F}_2^n$  has covering radius at most  $R$  when every word in  $\mathbb{F}_2^n$  is within Hamming distance  $R$  of some codeword. The covering number  $K_2(n, R)$  is the minimum possible size of such a code. Covering codes are a classical subject in coding theory, with connections to data compression, test design, combinatorial coverings, and domination in Hamming graphs [1, 3].

The binary covering-code table maintained by Keri records

$$15 \leq K_2(11, 3) \leq 16.$$

Its update log dates the lower-bound entry to 2006, while the table attributes the upper-bound line to the Graham–Sloane construction record [5, 3]. A targeted literature audit in September 2026 found no later primary source determining this exact cell. This dated audit is not a proof of absence. Recent work on semidefinite lower bounds also illustrates that a modern, proof-carrying treatment of individual table cells remains useful [2].

There are two definitive ways to close the gap. A verified 15-word cover would prove  $K_2(11, 3) = 15$ . A checked exclusion of every 15-word candidate, combined with a verified 16-word cover, would prove  $K_2(11, 3) = 16$ . The work reported here does neither. Its contribution is a finite and reproducible reduction of the second route.

| Item                                    | Status                       |
|-----------------------------------------|------------------------------|
| Known interval                          | $15 \leq K_2(11, 3) \leq 16$ |
| Verified 16-word radius-3 cover         | Yes                          |
| Verified 15-word radius-3 cover         | No                           |
| Complete normalized parent partition    | 150 branches                 |
| Certified normalized branch closures    | 112                          |
| Residual normalized branches            | 38                           |
| Selected fourth-word branches           | 350                          |
| Certified selected fourth-word closures | 324                          |
| Exact value determined                  | No                           |

Table 1: The certified frontier. The fourth-word layer is local to four selected children and does not reduce the global count of 38 residual normalized branches.

The reduction combines exact analytic inequalities, Hamming-cube automorphisms, orbit averaging, exact integer enumeration, and proof-producing satisfiability solving. Every retained SAT-based exclusion is tied to a reconstructed formula and a separately checked proof trace. The complete proof artifacts are archived with the version-specific DOI 10.5281/zenodo.22302261. The archived file and its SHA-256 digest are, respectively,

ruturajr-raval/binary-covering-code-11-3-v0.2.0.zip  
750003eba2e9f9baf5fee9ed93c679b3661daf6d8c68ca40eeb681202b5e72ff

After the source archive is authenticated against the SHA-256 published with the release, its compact replay checks the principal ledgers, the root and fourth-word proof manifests, and the Zenodo archive binding. It also regenerates the analytic certificates and verifies the known 16-word construction.

## 2 Definitions and the verified upper endpoint

**Definition 1.** For  $x, y \in \mathbb{F}_2^n$ , let  $d_H(x, y)$  be their Hamming distance. The radius- $R$  ball centered at  $x$  is

$$B_R(x) = \{y \in \mathbb{F}_2^n : d_H(x, y) \leq R\}.$$

A code  $C \subseteq \mathbb{F}_2^n$  is an  $(n, R)$  covering code when

$$\bigcup_{c \in C} B_R(c) = \mathbb{F}_2^n.$$

For  $(n, R) = (11, 3)$ , the ambient space has 2048 words and each ball has

$$|B_3(0)| = \sum_{i=0}^3 \binom{11}{i} = 232.$$

The repository contains the following 16-word linear code:

```
000000000000 00010001001 00101010101 00111011100
01001101011 01011100010 01100111110 01110110111
10001001000 10011000001 10100011101 10110010100
11000100011 11010101010 11101110110 11111111111.
```

Direct enumeration and a syndrome-space computation both give covering radius exactly 3. This verifies the known upper bound  $K_2(11, 3) \leq 16$ ; no originality is claimed for the construction.

### 3 Exact structural constraints at size 15

Assume throughout this section that  $C \subseteq \mathbb{F}_2^{11}$  is a radius-3 cover with  $|C| = 15$ . Let  $p_d$  be the number of unordered codeword pairs at Hamming distance  $d$ .

#### 3.1 Shell and Delsarte rows

There are 105 unordered pairs, so

$$\sum_{d=1}^{11} p_d = 105.$$

For a fixed codeword and a target shell at distance  $j$ , define

$$H(d, j) = \sum_{\substack{0 \leq a \leq d \\ 0 \leq j-a \leq 11-d \\ d+j-2a \leq 3}} \binom{d}{a} \binom{11-d}{j-a}.$$

Summing the coverage requirement over codewords and over every target in shell  $j$  gives, for  $4 \leq j \leq 11$ ,

$$2 \sum_{d=1}^{11} H(d, j) p_d \geq 15 \binom{11}{j}.$$

Let  $K_k(d)$  be the binary Krawtchouk polynomial. Fourier expansion of the distance distribution gives

$$15 \binom{11}{k} + 2 \sum_{d=1}^{11} K_k(d) p_d = \sum_{\text{wt}(a)=k} \left( \sum_{c \in C} (-1)^{a \cdot c} \right)^2.$$

Every inner sum is odd because  $|C| = 15$ . Its square is therefore at least 1, yielding the parity-refined inequality

$$2 \sum_{d=1}^{11} K_k(d) p_d \geq -14 \binom{11}{k}.$$

**Theorem 2** (Distance-distribution constraints). *Every 15-word radius-3 cover in  $\mathbb{F}_2^{11}$  satisfies*

$$\sum_{d=1}^6 p_d \geq 28, \quad \sum_{d=1}^5 p_d \geq 11.$$

*In particular, its minimum distance is at most 5.*

*Proof.* The first inequality is the exact nonnegative combination

$$\frac{1}{3}P + \frac{1}{24}D_1 + \frac{1}{24}D_{11},$$

where  $P$  is the pair-count equality and  $D_k$  is the degree- $k$  parity-refined Delsarte row. Its coefficient vector, ordered by distances 1 through 11, is

$$(1, 1, \frac{2}{3}, \frac{2}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, -\frac{1}{3}, -\frac{1}{3}, -\frac{2}{3}).$$

This is coefficientwise at most the indicator of distances 1 through 6, and the resulting rational lower bound is 28.

For the second inequality, combine

$$\frac{37}{615}P + \frac{1}{410}S_4 + \frac{11}{82}S_{11} + \frac{33}{820}D_1 + \frac{13}{2460}D_9,$$

where  $S_j$  denotes the shell- $j$  row. Coefficientwise comparison with  $p_1 + \dots + p_5$  gives the lower bound  $12467/1230$ . The left side is an integer, so it is at least 11. Both combinations and every coefficient are recomputed using exact rational arithmetic by the ancillary verifier.  $\square$

### 3.2 Pair and triple overlap

Let

$$O = \sum_{\{c, c'\} \subseteq C} |B_3(c) \cap B_3(c')|$$

be the total pair-ball overlap. The intersection sizes for pair distances  $1, \dots, 11$  are

$$112, 112, 56, 56, 20, 20, 0, 0, 0, 0, 0.$$

Normalize the shell-10, shell-11, degree-1 Delsarte, and degree-11 Delsarte rows by dividing their even coefficients by 2. Explicitly,

$$\begin{aligned} J_{10} &= (0, 0, 0, 0, 0, 0, 4, 3, 11, 11, 11) \cdot p \geq 83, \\ J_{11} &= (0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1) \cdot p \geq 8, \\ D_1 &= (9, 7, 5, 3, 1, -1, -3, -5, -7, -9, -11) \cdot p \geq -77, \\ D_{11} &= (-1, 1, -1, 1, -1, 1, -1, 1, -1, 1, -1) \cdot p \geq -7. \end{aligned}$$

Here  $p = (p_1, \dots, p_{11})$ . Exact coefficient comparison gives

$$\begin{aligned} O &= 20P + 4J_{10} + 4J_{11} + 9D_1 + 9D_{11} \\ &\quad + 20(p_1 + p_2) + 4(p_9 + p_{10}) + 40p_{11}. \end{aligned}$$

The row bounds give a base value 1708. Every coefficient of  $J_{10} + J_{11}$  is divisible by 4, while  $83 + 8 \equiv 3 \pmod{4}$ . The shell slacks  $u = J_{10} - 83$  and  $v = J_{11} - 8$  therefore satisfy  $u + v \equiv 1 \pmod{4}$ , hence  $u + v \geq 1$ .

**Theorem 3** (Overlap constraints). *Every 15-word radius-3 cover satisfies*

$$O \geq 1712 + 20(p_1 + p_2) + 4(p_9 + p_{10}) + 40p_{11}.$$

*In particular,  $O \geq 1712$ . If*

$$T = \sum_{\{c_1, c_2, c_3\} \subseteq C} |B_3(c_1) \cap B_3(c_2) \cap B_3(c_3)|,$$

*then  $T \geq 280$ .*

*Proof.* The displayed exact identity and the shell-slack congruence prove the pair-overlap inequality. For the triple term, let  $m_x$  be the number of codeword balls containing  $x$ . Since  $C$  is a cover,  $m_x \geq 1$ . Pointwise,

$$\binom{m_x}{3} \geq \binom{m_x}{2} - (m_x - 1).$$

The total incidence excess is

$$15 \cdot 232 - 2048 = 1432.$$

After summing over  $x \in \mathbb{F}_2^{11}$ , we obtain  $T \geq O - 1432 \geq 280$ .  $\square$

## 4 The size-15 formula

Introduce one Boolean variable  $x_v$  for each  $v \in \mathbb{F}_2^{11}$ . Coverage is expressed by the 2048 clauses

$$\bigvee_{d_H(u,v) \leq 3} x_v, \quad u \in \mathbb{F}_2^{11}.$$

An exact or at-most-15 cardinality constraint restricts the selected codewords. For a nonempty covering code, an at-most-15 model can be padded to exactly 15 distinct codewords without losing coverage, so the two formulations are equivalent for existence.

Two separately audited encodings were retained. A sequential exact-cardinality formula has 34816 variables and 131029 clauses. A compact at-most-15 totalizer formula has 9464 variables and 26804 clauses. Projection tests on smaller instances verify that auxiliary variables preserve the intended codeword semantics.

The Hamming cube automorphism group contains all translations and coordinate permutations. The reduction uses only transformations that are explicitly shown to preserve coverage and the formula assumptions. No solver-discovered symmetry-breaking condition enters the proof ledger without a separate completeness argument.

## 5 A complete 150-case normalized partition

Since  $15 \cdot 232 > 2048$ , the 15 radius-3 balls cannot be pairwise disjoint. Thus some pair of codewords has distance at most 6. Choose a globally closest pair, let its distance be  $w$ , and translate one endpoint to zero. The other endpoint is then a minimum-weight nonzero codeword of the translated code. A coordinate permutation maps it to

$$A = 2^w - 1, \quad 1 \leq w \leq 6.$$

The stabilizer of  $A$  is  $S_w \times S_{11-w}$ . Its orbit on a second word  $B$  is determined by

$$(\text{wt}(B), |\text{supp}(A) \cap \text{supp}(B)|).$$

Choose the lexicographically first occupied orbit and map one selected word in it to its canonical representative. Across the six possible minimum weights, this produces 150 canonical parent branches.

**Theorem 4** (Certified normalized frontier). *If a 15-word radius-3 cover exists, a Hamming-cube automorphism and the documented closest-pair and maximum-degree normalizations map it into one of 38 explicitly retained normalized branches.*

*Proof.* The complete two-word orbit manifest contains 150 branches. The stage-1 ledger classifies them in the disjoint precedence order shown in Table 2. Exact Farkas multipliers close 80 branches. Exhaustive integer orbit-profile certificates close two more. One standalone formula and 14 closest-pair formulas have checked DRAT proofs. Theorem 2 closes the four remaining weight-6 branches in the broader first-stage ledger.

The 49 stage-1 residual branches are then normalized as follows. For a code of minimum distance  $d$ , form the graph on its codewords in which two vertices are adjacent exactly when their distance is  $d$ . Choose a maximum-degree vertex, translate it to zero, choose one of its neighbors, and map that neighbor to  $A = 2^d - 1$ . This is compatible with the two-word partition because translations and coordinate permutations preserve distances and coverage.

Let  $B$  be the selected word in the first remaining descriptor. If  $\text{wt}(B) = d$ , then zero has at least two minimum-distance neighbors, and no additional conclusion is drawn. If  $\text{wt}(B) > d$ , the descriptor ordering implies that  $A$  is the only selected word of weight  $d$ , so zero has degree one. If  $d_H(A, B) = d$ , then  $A$  has at least the two neighbors zero and  $B$ , contradicting the maximum-degree choice. This excludes five branches. If instead  $d_H(A, B) > d$ , the maximum degree is one and the entire minimum-distance graph is a matching. This proves the completeness of the classification into 5 contradiction, 34 matching, and 10 multiple-neighbor cases.

Six complete third-word formulas have checked DRAT proofs, four within the separately proved matching scope and two without that condition. The machine ledger checks disjointness, all 150 identities, all input hashes, and the final count  $150 - 112 = 38$ .  $\square$

| Closure reason                             | Branches |
|--------------------------------------------|----------|
| Exact rational orbit-profile infeasibility | 80       |
| Exact integer orbit-profile infeasibility  | 2        |
| Standalone checked DRAT                    | 1        |
| Minimum distance at most 5                 | 4        |
| Closest-pair checked DRAT                  | 14       |
| Maximum-degree contradiction               | 5        |
| Checked third-word DRAT                    | 6        |
| Certified normalized closures              | 112      |
| Residual normalized branches               | 38       |

Table 2: The complete normalized parent ledger. The advanced rows are normalization statements and must not be reinterpreted as 11 additional unrestricted parent-CNF exclusions.

*Remark 5.* The exact rational orbit analysis leaves 70 cases feasible over the reals. Integer orbit counts exclude two of these, leaving 68 feasible aggregate profiles. Such profiles are necessary averaged data, not vertex-level covering codes.

## 6 Third-word and fourth-word orbit frontiers

Fix a canonical parent with selected words  $0, A, B$ . The ordered pair  $(A, B)$  partitions the 11 coordinates into four cells:

$$A \cap B, \quad A \setminus B, \quad B \setminus A, \quad [11] \setminus (A \cup B).$$

The pointwise stabilizer permutes coordinates within these cells. Therefore the orbit of a third word  $D$  is determined by its four intersection counts. Selecting the first occupied orbit gives an exhaustive child split.

The 49 stage-1 residual parents have 3238 such child orbits. Six third-word DRAT closures remove 423 children, and five maximum-degree-closed parents account for 267 more. The 38 active parents contain 2548 raw children. Existing minimum-distance clauses exclude 312, and the certified matching condition excludes 73 additional children in its valid scope. Exactly 2163 live children remain.

The next retained layer studies four selected unresolved children below the parent `w4-weight5-intersection0`. Once three nonzero words are fixed, the coordinates split into eight membership

cells. A fourth word is classified by its eight intersection counts. In each selected child, the first occupied fourth-word orbit gives an exhaustive split because the fixed words leave more than 1200 ambient vertices uncovered.

**Theorem 6** (Selected fourth-word reduction). *The four selected third-word children in Table 3 split into 350 exhaustive fourth-word branches. Checked reverse unit propagation certificates close 184 branches, and checked solver-generated DRAT certificates close 140 more. Exactly 26 branches remain unresolved. Every selected child retains at least one residual branch.*

*Proof.* The separate orbit auditor reconstructs the eight coordinate cells, all candidate words, static exclusions, canonical representatives, and branch digests. It obtains 350 branches.

For 184 formulas, the branch units force a contradiction by reverse unit propagation. Each resulting trace is checked against the reconstructed formula. The remaining 166 formulas formed the exploratory solver frontier. A retained proof-producing rerun generated DRAT traces for 140 of them, while 26 remained unresolved. Each retained trace is checked by `drat-trim` at pinned revision `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. The proof indexes authenticate every formula, trace, checker result, and production record. Clean-checkout self-attestation replay passed for both layers. The per-child residual counts in Table 3 sum to 26.  $\square$

| Child     | RUP | DRAT | Residual | Total |
|-----------|-----|------|----------|-------|
| orbit-005 | 50  | 29   | 6        | 85    |
| orbit-007 | 53  | 15   | 8        | 76    |
| orbit-014 | 41  | 28   | 4        | 73    |
| orbit-015 | 40  | 68   | 8        | 116   |
| Total     | 184 | 140  | 26       | 350   |

Table 3: Certified branch-level results inside four selected unresolved third-word children.

**Corollary 7.** *The selected fourth-word layer closes no complete third-word child and no normalized parent. Consequently, it does not reduce the global count of 38 residual normalized branches.*

## 7 Certificate architecture and reproduction

The proof records separate mathematical scope from computational status. A timeout, an unlogged UNSAT response, or a heuristic search failure is never entered as an exclusion. A retained SAT-based closure contains:

1. an exact formula identity and semantic metadata;
2. a deterministic formula reconstruction path;
3. a SHA-256 digest of the formula;
4. a compressed RUP or DRAT trace;
5. a checker record tied to a pinned `drat-trim` revision;
6. an exact-membership manifest for the proof bundle; and

7. a clean-checkout replay record.

The DRAT format permits compact separately checked certificates for SAT-solver refutations [4].

The arXiv source archive intentionally omits the large proof payloads. It contains the exact ledgers, proof indexes, replay records, the 16-word code, and a deterministic manifest. Running

```
python3 replay.py
```

from the extracted source:

1. checks every ancillary file against the internal manifest;
2. verifies the 16-word covering radius by direct enumeration;
3. checks the recorded 150-to-38 and 350-to-26 ledger counts, statuses, and source digests;
4. checks the retained RUP and solver-DRAT proof-index identities;
5. checks the retained clean-checkout replay statuses; and
6. regenerates the distance and overlap certificates byte-for-byte using standard-library Python.

No network access or third-party package is required for this compact replay. Full replay of all 324 retained proof traces requires the immutable v0.2.0 artifact deposit at 10.5281/zenodo.22302261 and the pinned verification environment. The all-versions concept DOI is 10.5281/zenodo.22260709. The paper-inclusive archival release is version v0.3.1, DOI 10.5281/zenodo.22647766. This archival and documentation patch adds an explicit compiled PDF and deterministic source bundle; the theorem, proof certificates, retained data, and computations are unchanged.

## 8 Limitations and next steps

The known interval remains

$$15 \leq K_2(11, 3) \leq 16.$$

The present work does not prove a new lower bound and does not produce a 15-word cover. Thirty-eight normalized parent branches remain unresolved. The 2163 live third-word children have not been exhaustively decided. The selected fourth-word bundle leaves 26 branches and closes no complete selected child.

The immediate proof route is an audited fifth-word orbit split of those 26 branches. Closing all four selected children would still leave other live children below the 38 residual parents, so subsequent work must expand the certified cover rather than extrapolate from local success. In parallel, nonlinear construction search remains valuable: a single independently verified 15-word cover would settle the problem in the opposite direction.

The methods are reusable beyond this table cell. Exact orbit averaging, integer profile certificates, stabilizer-aware SAT formulas, and proof-logged branch exclusions apply to finite covering and domination problems with large automorphism groups.

## 9 Conclusion

We have converted a broad size-15 search into an explicit certified frontier. Every hypothetical cover can be normalized into one of 38 retained branches. Within four selected unresolved children, 324 of 350 fourth-word branches have checked certificates. These are substantial branch-level reductions, but they are not a solution of  $K_2(11, 3)$ . The remaining cases are stated explicitly so that future computation can extend a verified proof cover rather than repeat an unauthenticated search.



## References

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